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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Sub'ect	DSA-2
Academic Year	1-III 2025-26 Semester
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Course Outcome	CO-2

Case Study Topic: Tournament Statistics Analysis using Segment Tree

Work Done Summary:

The SportsAnalytics system stores scores of multiple matches played in a tournament. Coaches and organizers frequently need information such as total runs scored between specific matches and highest scores in a range. A Segment Tree is implemented to answer range queries efficiently.

Implementation Details:

1. Store match scores in an array.
2. Construct a Segment Tree from the scores.
3. Perform range sum queries.
4. Update match scores whenever corrections are made.
5. Analyze performance compared to linear searching.

Result: Screenshots / Output

```

1  class SegmentTree {
2      int[] tree;
3      int n;
4      SegmentTree(int arr[]) {
5          n = arr.length;
6          tree = new int[4 * n];
7          build(arr, 1, 0, n - 1);
8      }
9      void build(int arr[], int node, int start, int end) {
10         if (start == end) {
11             tree[node] = arr[start];
12         } else {
13             int mid = (start + end) / 2;
14
15             build(arr, 2 * node, start, mid);
16             build(arr, 2 * node + 1, mid + 1, end);
17
18             tree[node] = tree[2 * node] + tree[2 * node + 1];
19         }
20     }

```

```

20     }
21     int query(int node, int start, int end, int l, int r) {
22         if (r < start || end < l)
23             return 0;
24         if (l <= start && end <= r)
25             return tree[node];
26         int mid = (start + end) / 2;
27         return query(2 * node, start, mid, l, r)
28             + query(2 * node + 1, mid + 1, end, l, r);
29     }
30 }
31 public class SportsAnalyticsSegmentTree {
32     public static void main(String[] args) {
33         int scores[] = {120, 140, 110, 180, 200, 160, 130};
34         SegmentTree st = new SegmentTree(scores);
35         System.out.println("Total Score from Match 2 to Match 6 :
36             + st.query(1, 0, scores.length - 1, 1, 5));
37     }
38 }

```

Segment Tree successfully performed range queries in $O(\log n)$ time and improved tournament statistics retrieval efficiency.

GitHub Link: <https://github.com/Shiva77661/SportsAnalytics-Athlete-Performance-Tournament-Management-System/commit/15fb54c74c325ee1483736796473fc7c03aa19c4>

Student Signature

Faculty Signature